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MILITARY SPECIFICATION

DAMPENERS, FLUID PRESSURE, GAGE PROTECTION

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. Fluid pressure dampeners covered by this specification are for use on gage lines and include but are not limited to devices commercially known as pulsation dampeners, pressure snubbers, pressure equalizers, and gage savers.

1.2 Classification. Fluid pressure dampeners covered by this specification shall be of the following types, classes, and compositions, as specified (see 6.2):

Type I — Oil service (above 225 Saybolt Seconds. Universal (SSU) at operating temperature).

Type II — Gasoline, fresh water (light oil (below 225 SSU) or steam).

Type III — Gases.

Type IV — Sea water (composition b only).

Class 1 — 0 to 800 pounds per square inch (p. s. i.) working pressure.

Class 2 — 801 to 5,000 p.s.i. working pressure.

Class 3 — 5,001 to 15,000 p.s.i. working pressure.

Composition a — Brass or bronze.

Composition b — Nickel-copper alloy.

Composition c — Corrosion-resisting steel.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-B-621 — Brass, Castings, Leaded Yellow.

QQ-B-626 — Brass, Leaded and Nonleaded Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips).

FSC 6685

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QQ-M-80 — Manganese Bronze Bars, Plates, Rods, Sheets, Strips, Flat Wire, Forgings, and Structural and Special Shaped Sections.

QQ-N-281 — Nickel - Copper - Alloy (Monel and R-Monel) Bars Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.

QQ-S-763 — Steel Bars, Shapes and Forgings—Corrosion Resisting.

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-10377 — Boxes: Wood-Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

MIL-N-17163 — Nickel-Copper Alloy, Wrought; (55 to 50 percent Nickel) Low Permeability.

MIL-G-18997 — Gages, Pressure, Dial Indicating, Bourdon Tube.

STANDARDS

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York, N. Y.)

NATIONAL BUREAU OF STANDARDS

Handbook H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Qualification. Fluid pressure dampeners, furnished under this specification, shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3).

3.2 Material. The materials for construction of the bodies of dampeners, composi-

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tions a, b, and c, shall be in accordance with specifications referenced herein or of commercial material having an equivalent chemical composition. Internal parts shall be of materials equal or superior in corrosion resistance to that used in the body, but corrosion-resistant steel is not acceptable for type IV dampeners.

3.2.1 *Composition a.* The material for composition a shall be either bronze or brass in accordance with Specification QQ–M–80, QQ–B–626 or QQ–B–621.

3.2.2 *Composition b.* The materials for composition b shall be nickel-copper alloy in accordance with Specification MIL–N–17163 or QQ–N–281.

3.2.3 *Composition c.* The materials for composition c shall be corrosion-resisting steel in accordance with Specification QQ–S–763.

3.3 **Design.** The design of fluid pressure dampeners shall be as follows:

3.3.1 *Construction.* The length of the body of the dampeners shall not exceed 8 inches. The body shall have hexagonal cross-sections of sufficient widths to accommodate the use of a wrench(s) for disassembling, installing and removing.

3.3.2 *End connections.* The end connections of the dampeners shall be externally threaded on one end and internally threaded on the other end. The size of the end connections shall be ¼ inch American Standard Pipe threads, in accordance with Handbook H28. Both end connections shall be of the same size.

3.3.3 *Assembly.* When the dampener design includes moving parts or assembled parts, it shall be so designed that the internal parts shall be positively locked into place preventing the parts from becoming loose and inadverently dropping into the pipeline. Split pin type of retaining pins are not acceptable. Porous metal type dampeners are not acceptable for type III.

3.4 **Performance.** The performance shall be as follows:

3.4.1 *Operation.* The operation of the dampener is to protect gages against sudden changes in pressure which cause fluctuation of the pointer or hand, resulting in inaccuracy of reading or damage to the gage. Correction of gage calibration accuracy by this device is not expected.

3.4.2 *Dampening.* The total excursion of the dampened pulsation shall be less than one scale subdivision of the gage when the undampened excursion is 60 percent of the scale range at a frequency of one cycle per second (c.p.s.) in accordance with 4.4.1.

3.4.3 *Response.* When the pressure applied to the gage with pulsation dampener is suddenly changed from 20 to 80 percent at the dial range, the gage shall indicate 58 percent of the dial range within 30 seconds, with a volume of 7 cubic inches between dampener and gage.

3.4.4 *Accuracy.* The mean pressure indicated by the gage used with a pulsation dampener under test shall not differ by more than one-half percent from the mean pressure indicated by a duplicate gage used with a previously laboratory approved pulsation dampener when both are connected to a pipe line carrying a medium under pulsating pressure, as specified in 3.4.2. The accuracy of the gage shall not be affected by the dampener.

3.4.5 *Direction of flow.* The direction of flow through the fluid pressure dampener shall be indicated by an arrow with the word "Gage" at the point of the arrow.

3.5 **Marking.** Each dampener shall be stamped or otherwise permanently marked

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with the manufacturer's stamp or trademark, type, class and composition.

3.6 Workmanship. Fluid pressure dampeners shall be free from all defects that may affect their serviceability and shall be of the best commercial quality. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling.

4.2.1 *Lot.* A lot shall consist of all dampeners of the same type, class and composition offered for delivery at one time.

4.2.2 *Sampling for visual and dimensional examination and strength test.* A random sample of dampeners shall be selected by the Government inspector in accordance with table I from each lot of material offered for Government examination of visual and dimensional characteristics and the test of 4.4.2 with lot acceptance based on the sampling requirements of table I in accordance with Standard MIL–STD–105.

4.2.3 *Sampling for test.* A random sample of dampeners shall be selected in accordance with table II from each lot for the test specified in 4.3.2. Any sample failing to pass the test shall be rejected, and if the number of rejected dampeners in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE I. *Sampling for visual and dimensional examination and strength test*
AQL (approx.) = 1.5 percent defective

Number of fluid pressure dampeners in inspection lot	Number of fluid pressure dampeners in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

TABLE II. *Sampling for acceptance tests.*

Number of fluid pressure dampeners in inspection lot	Number of fluid pressure dampeners in the sample	Acceptance number	Rejection number
		(Number of items nonconforming on any acceptance test)	
40 and under	1	0	0
41 to 110	2	0	1
111 to 300	3	0	1
301 to 800	4	0	1
801 to 1300	5	0	1
1301 and over	7	0	1

4.3 Acceptance inspection (at place of manufacture).

4.3.1 *Visual and dimensional examination.* Each of the sample dampeners selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any dampener in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective dampeners in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Acceptance test.* Each of the samples selected in accordance with 4.2.3 shall be subjected to the test specified in 4.4.1.

4.4 Tests.

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4.4.1 *Response.* Sample pulsation dampeners of all types, classes, and compositions shall be given the following response test: Two 6-inch-dial gages, having the proper bourdon tube material and correct scale range, in accordance with Specification MIL–G–18997, shall be selected upon which the test shall be performed. The bourdon tubes of the gages to be dampened shall be evacuated by a vacuum pump and at the same time filled solid with the fluid to be used, before attaching the dampener sample. *(Note.*—Evacuation of air from the tubes increases the responsiveness of the gages.) The two gages shall be connected on a whip test manifold specified in Specification MIL–G–18997. The ratio of the pulsation shown on the dampened gage to the pulsation shown on the undampened gage shall be computed. The control valve shall be cycled and time observed and recorded for the dampened gage to indicate a change of pressure between 20 to 80 percent of the total dial range of the gage, in not more than one minute or not less than one second. (See 3.4.2, 3.4.3 and 3.4.4.)

4.4.2 *Strength.* The dampeners shall be subjected to a hydrostatic pressure of 150 percent of the maximum rated pressure (see 1.2), for a period of 10 minutes without damage or evidence of leakage.

4.5 **Qualification tests (at a laboratory).** Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.4.

4.6 Rejected lots may be resubmitted after reworking and removing nonconforming units from the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The fluid pressure dampeners shall be individually preserved and packaged in accordance with method III of Specification MIL–P–116. Contact preservative is not required.

5.1.2 *Level C.* Fluid pressure dampeners shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Fluid pressure dampeners, packaged in accordance with level A or C as specified (see 6.2) shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood cleated paper overlaid, or wood cleated plywood boxes conforming to Specification PPP–B–591, PPP–B–621, PPP–B–585, PPP–B–636 (class 3), MIL–B–10377 or PPP–B–601, respectively, at the option of the contractor. Shipping containers shall have case liners conforming to Specification MIL–L–10547 and shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to Specification PPP–B–636 may be omitted provided all joints of the boxes are sealed with tape conforming to Specification PPP–T–76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 *Level B.* Fluid pressure dampeners, packaged in accordance with level A or C as specified (see 6.2) shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood cleated plywood or wood cleated paper overlaid boxes conforming to Specification PPP–B–591, PPP–B–621, PPP–B–585, PPP–B–636 (class 2), PPP–B–601, or MIL–B–10377, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification

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or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.3 *Level C.* Fluid pressure dampeners, packaged for level A or C, as specified (see 6.2) shall be packed in containers of the type, size and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking required by the contract, or order, or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** Fluid pressure dampeners are intended for use in the following systems:

Type
Oil
Fresh water, sea water, steam, light oil
Gasoline
Gases

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, class, and composition of fluid pressure dampeners (see 1.2).

(c) Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL–2940, whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

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